

WASP-36b

R-0865

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_230108 · created 2026-07-28T22:35:32+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459952.8601 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.048
Transit depth (Rp/Rs)^2	2.62 +/- 1.54 [%]
Orbital Inclination (inc)	87.79 +/- 2.88
Airmass coefficient 1 (a1)	4.47 +/- 2.7
Airmass coefficient 2 (a2)	-1.1 +/- 1.07
Scatter in the residuals of the lightcurve fit is	58.81 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.092 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.048 vs 0.1368 (deviation 0.53σ)
Duration fit vs published	0.092 d vs 0.0758 d (deviation 1.62σ)
Mid-time O–C	-10.3 min
Depth SNR	3.4
Residual scatter	58.81 %

Light curve

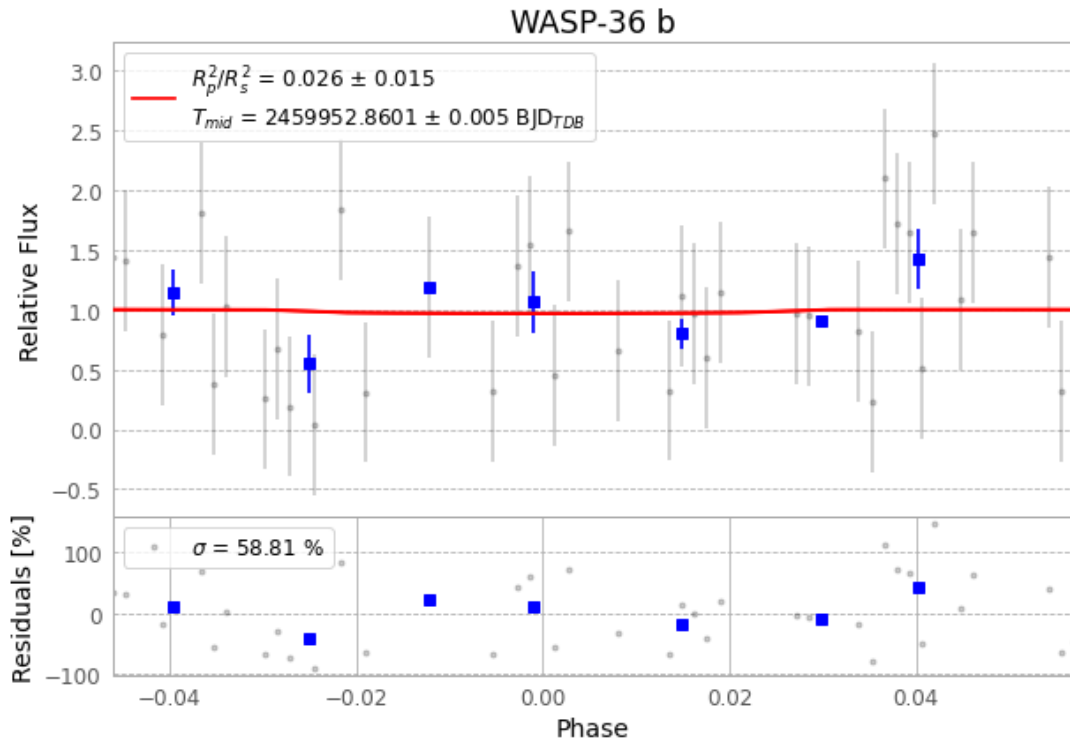


Figure 1 — FinalLightCurve_WASP-36 b_2023-01-07.png (SHA-256 d3773f8cf8ebfad8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [561, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bfa5c5e1ea2c16cd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36230108064815.FITS → WASP-36230108103615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-230108.log (f0c1c8f9ebe5...), FinalLightCurve_WASP-36 b_2023-01-07.png (d3773f8cf8eb...), FinalLightCurve_WASP-36 b_2023-01-07.csv (612f922c3337...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 22:35Z.